

Lecture 14

Interpolation of Spatial Data I

STAT 8020 Statistical Methods II

Background

Gaussian Process
Spatial Model

Spatial Interpolation

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Gaussian Process
Spatial Model

Spatial Interpolation

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2 Gaussian Process Spatial Model

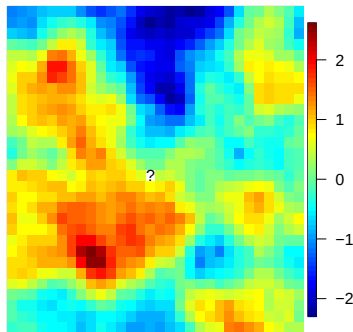
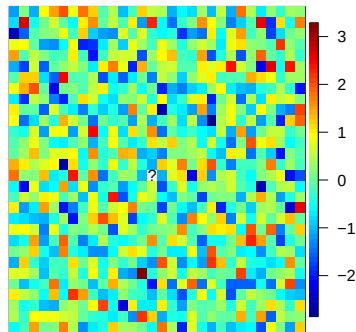
3 Spatial Interpolation

By the end of this module, you should be able to:

- Explain how spatial dependence can be exploited for interpolation and prediction
- Formulate Gaussian process models for spatially referenced data
- Understand the concepts of stationarity and isotropy and their implications for covariance modeling
- Identify commonly used covariance functions and characterize their smoothness and dependence properties
- Explain the principles of kriging as the best linear unbiased predictor
- Implement Gaussian process estimation, perform kriging prediction, and interpret associated uncertainty maps

Toy Examples of Spatial Interpolation

Consider two spatial images, each with a missing pixel denoted by “?”



Question: What is your best guess of the value of the missing pixel, denoted as $Y(s_0)$, for each case?

Discussion

Which image allows a more accurate prediction? Why? What information are you using?

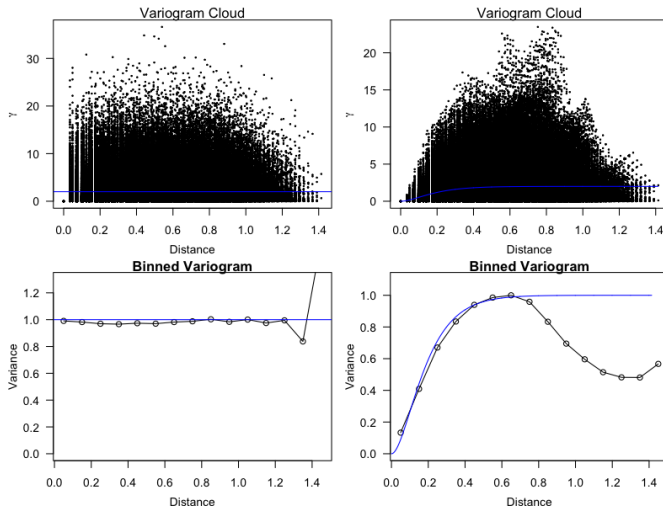
Visualizing Spatial Dependence Structure

Closer locations are more alike; the **variogram** measures how similarity decays with distance $\gamma(h) = \frac{1}{2} \text{Var}(Y(s) - Y(s+h))$

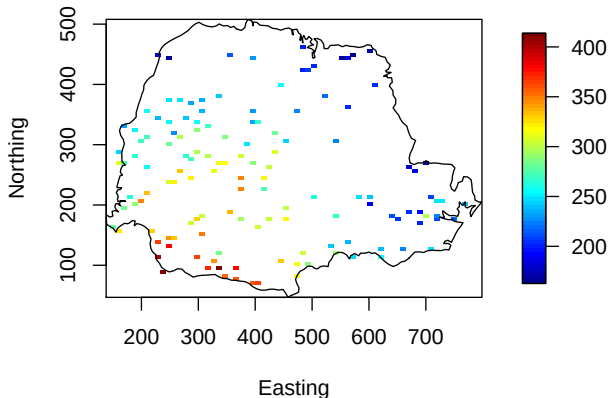
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Interpolating Paraná State Precipitation Data



Challenge

Measurements are only available at monitoring stations.
How can we estimate precipitation everywhere else?

The Spatial Interpolation Problem

Given observations of a spatially varying quantity Y at n spatial locations

$$y(s_1), y(s_2), \dots, y(s_n), \quad s_i \in \mathcal{S}, i = 1, \dots, n$$

We want to estimate this quantity at any **unobserved location**

$$Y(s_0), \quad s_0 \in \mathcal{S}, \quad \hat{Y}(s_0) = E[Y(s_0) \mid \mathbf{Y} = \mathbf{y}]$$

⇒ The optimal predictor under **squared error loss** is the **conditional expectation**

Applications

- Mining: ore grade
- Climate: temperature, precipitation, ...
- Environmental Science: air pollution levels, ...

A Brief History of Spatial Statistics

● Applied Origins

- Mining [Krige, 1951], geology, and forestry: interpolation and resource estimation



● Geostatistics

- Variograms [Matheron, 1960s]. Matérn (1960): covariance models



● Statistical Foundations

- Cressie (1993)
- Stein (1999)



Spatial statistics evolved from geoscience applications to a rigorous statistical discipline based on stochastic processes, covariance modeling, and spatial prediction

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Linear Interpolation

The best guess (under squared loss) should be based on the conditional distribution $[Y(s_0) | Y = \mathbf{y}]$ where

$$\mathbf{y} = (y(s_1), \dots, y(s_n))^T$$

- Calculating this conditional distribution can be difficult
- Instead we use a **linear predictor**:

$$\hat{Y}(s_0) = \lambda_0 + \sum_{i=1}^n \lambda_i y(s_i)$$

- The best linear predictor is completely determined by the **mean** and **covariance** of $\{Y(s), s \in \mathcal{S}\}$

Why Linear?

Linear predictors are

- Computationally simple
- Optimal for Gaussian processes

Gaussian Process (GP) Spatial Model

Suppose the observed data $\{Y(\mathbf{s}_i)\}_{i=1}^n$ are measurements from an underlying spatial process

$$\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}},$$

where $\mathcal{S} \subset \mathbb{R}^d$ denotes the spatial domain.

Model: $Y(\mathbf{s}) = m(\mathbf{s}) + \epsilon(\mathbf{s}), \quad \mathbf{s} \in \mathcal{S}.$

- Mean function (trend):

$$m(\mathbf{s}) = \mathbb{E}[Y(\mathbf{s})] = \mathbf{x}^T(\mathbf{s})\boldsymbol{\beta},$$

which captures the large-scale spatial variation through spatial covariates

- Spatial residual process:

$$\{\epsilon(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}} \sim \text{GP}(0, K(\cdot, \cdot)),$$

where

$$K(\mathbf{s}_1, \mathbf{s}_2) = \text{Cov}(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2))$$

is the covariance function describing spatial dependence

Covariance Modeling Assumptions: Stationarity and Isotropy

Residual process:

$$\epsilon(\mathbf{s}) \sim GP(0, K).$$

To estimate covariance K from the observed data, we typically impose stationarity and isotropy assumptions

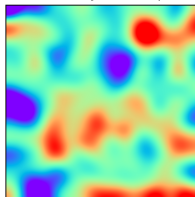
- Stationarity:

$$\begin{aligned} K(\mathbf{s}_1, \mathbf{s}_2) &= \text{Cov}(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2)) = C(\mathbf{s}_1 - \mathbf{s}_2) \\ &= \text{Cov}(\epsilon(\mathbf{s}_1 + \mathbf{h}), \epsilon(\mathbf{s}_2 + \mathbf{h})) \end{aligned}$$

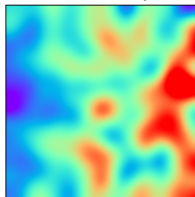
- Isotropy:

$$K(\mathbf{s}_1, \mathbf{s}_2) = \text{Cov}(\epsilon(\mathbf{s}_1), \epsilon(\mathbf{s}_2)) = C(\|\mathbf{s}_1 - \mathbf{s}_2\|)$$

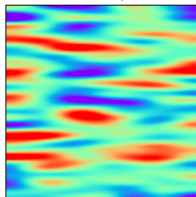
Stationary & Isotropic



Nonstationary



Anisotropic



A Valid Covariance Function Must Be Positive Definite!

A covariance function is positive definite (p.d.) if

$$\sum_{i,j=1}^n a_i a_j C(\mathbf{s}_i - \mathbf{s}_j) \geq 0$$

for any finite locations $\mathbf{s}_1, \dots, \mathbf{s}_n$, and for any constants a_i ,
 $i = 1, \dots, n$

Question: what is the consequence if a covariance function is NOT p.d.? $\Rightarrow$ **We can get a negative variance**

Question: How to guarantee a $C(\cdot)$ is p.d.?

- Using a **parametric covariance function** (see some examples in next slide)
- Using **Bochner's Theorem**  to construct a valid covariance function

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Some Commonly Used Covariance Functions

Let $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$ denotes the distance between two locations, σ^2 controls the marginal variance, and ρ controls the spatial range (how quickly dependence decays with distance)

- **Powered Exponential**

$$C(h) = \sigma^2 \exp\left\{-\left(\frac{h}{\rho}\right)^\alpha\right\}, \quad 0 < \alpha \leq 2.$$

- α controls smoothness
- $\alpha = 1$: Exponential covariance
- $\alpha = 2$: Gaussian covariance

- **Spherical**

$$C(h) = \sigma^2 \left(1 - \frac{3}{2} \frac{h}{\rho} + \frac{1}{2} \left(\frac{h}{\rho}\right)^3\right) 1_{\{h \leq \rho\}}.$$

- Compact support: covariance becomes exactly zero for $h > \rho$.
- Valid only for spatial dimensions $d = 1, 2, 3$

$$C(h) = \sigma^2 \frac{(\sqrt{2\nu}h/\rho)^\nu \mathcal{K}_\nu(\sqrt{2\nu}h/\rho)}{\Gamma(\nu)2^{\nu-1}}, \quad \sigma^2 > 0, \rho > 0, \nu > 0,$$

where

- ν : smoothness parameter
- $K_\nu(\cdot)$: modified Bessel function of the second kind

Why is the Matérn family popular?

- The smoothness parameter ν controls the differentiability of the spatial process.
- Exponential covariance when $\nu = 1/2$; Gaussian covariance as $\nu \rightarrow \infty$.
- The smoothness parameter ν allows more flexible modeling of spatial smoothness than the powered exponential model

“Use the Matérn model” – Stein (1999, pp. 14)

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1-D Realizations from Matérn Model with Fixed σ^2, ρ

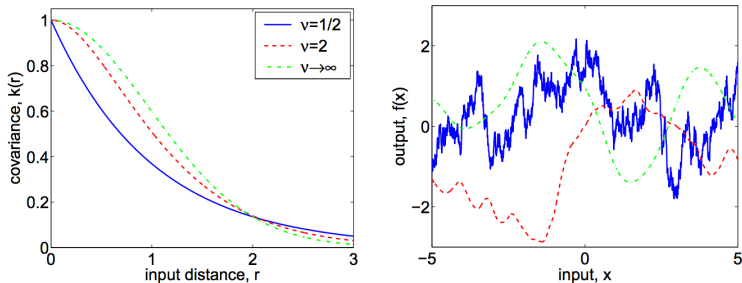


Figure: courtesy of Rasmussen & Williams 2006

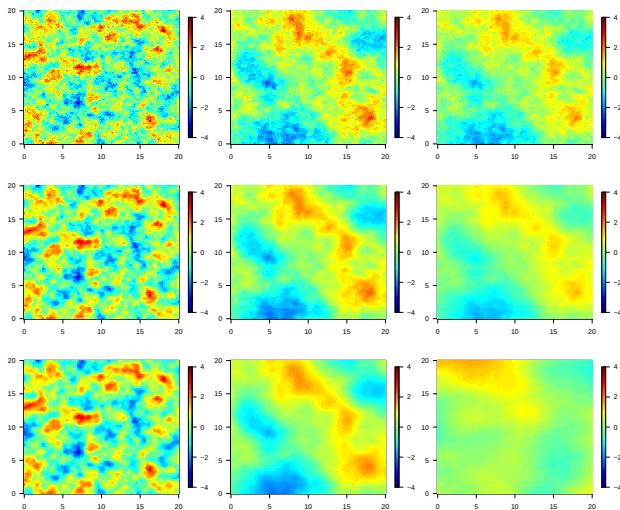
The larger ν is, the smoother the process is

2-D Realizations from Matérn Model with Fixed σ^2

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Left $\rightarrow$ Right: increasing range (ρ)
smoothness (ν) σ^2 fixed

Top $\rightarrow$ Bottom: increasing

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If

$$\begin{pmatrix} \mathbf{Y}_1 \\ \mathbf{Y}_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

Then

$$[\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2] \sim N(\boldsymbol{\mu}_{1|2}, \Sigma_{1|2})$$

where

$$\boldsymbol{\mu}_{1|2} = \boldsymbol{\mu}_1 + \Sigma_{12} \Sigma_{22}^{-1} (\mathbf{y}_2 - \boldsymbol{\mu}_2)$$

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}$$

Main Idea

Observed values provide information about unobserved values through the covariance structure

If $\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}}$ follows a GP, then

$$\begin{pmatrix} Y_0 \\ \mathbf{Y} \end{pmatrix} \sim N \left(\begin{pmatrix} m_0 \\ \mathbf{m} \end{pmatrix}, \begin{pmatrix} \sigma_0^2 & k^T \\ k & \Sigma \end{pmatrix} \right)$$

We have

$$[Y_0 | \mathbf{Y} = \mathbf{y}] \sim N(m_{Y_0 | \mathbf{Y} = \mathbf{y}}, \sigma_{Y_0 | \mathbf{Y} = \mathbf{y}}^2)$$

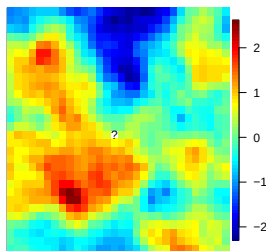
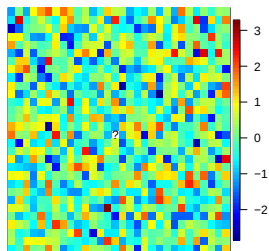
where

$$\begin{aligned} m_{Y_0 | \mathbf{Y} = \mathbf{y}} &= m_0 + k^T \Sigma^{-1} (\mathbf{y} - \mathbf{m}) \\ \sigma_{Y_0 | \mathbf{Y} = \mathbf{y}}^2 &= \sigma_0^2 - k^T \Sigma^{-1} k \end{aligned}$$

Next, we are going to revisit our toy examples

Toy Examples Revisited

For simplicity, we assume $m(s) = 0$ for $s \in \mathcal{S}$, the spatial covariance only depends on distance



$$m_{Y_0|Y=y} = 0 + k^T \Sigma^{-1} (y - 0), \quad \sigma_{Y_0|Y=y}^2 = \sigma_0^2 - k^T \Sigma^{-1} k$$

Spatial uncorrelated field:

- $m_{Y_0|Y} = 0$
- $\sigma_{Y_0|Y=y}^2 = \sigma_0^2$

Spatial correlated field:

- $m_{Y_0|Y} = k^T \Sigma^{-1} y$
- $\sigma_{Y_0|Y=y}^2 = \sigma_0^2 - k^T \Sigma^{-1} k$

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Interpolating Multiple Points in Space

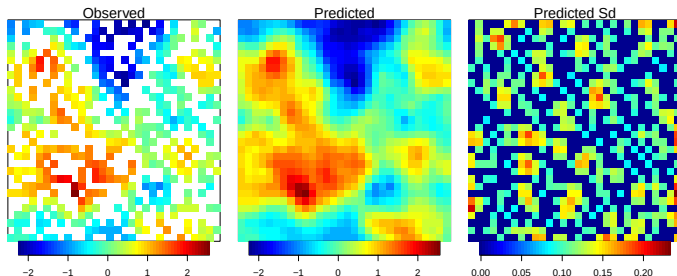
In practice, we would like to predict the values at many locations. The Gaussian conditional distribution formula can still be used:

$$[Y_0|Y = y] \sim N(m_{Y_0|Y=y}, \Sigma_{Y_0|Y=y})$$

where

$$m_{Y_0|Y=y} = m_0 + k^T \Sigma^{-1} (y - m)$$

$$\Sigma_{Y_0|Y=y} = \Sigma_0 - k^T \Sigma^{-1} k$$



If $\{Y(\mathbf{s})\}_{\mathbf{s} \in \mathcal{S}}$ follows a GP, then

$$\begin{pmatrix} Y_0 \\ \mathbf{Y} \end{pmatrix} \sim N \left(\begin{pmatrix} \mathbf{m}_0 \\ \mathbf{m} \end{pmatrix}, \begin{pmatrix} \Sigma_0 & \mathbf{k}^T \\ \mathbf{k} & \Sigma \end{pmatrix} \right)$$

We have

$$[Y_0 | \mathbf{Y} = \mathbf{y}] \sim N(\mathbf{m}_{Y_0 | \mathbf{Y} = \mathbf{y}}, \Sigma_{Y_0 | \mathbf{Y} = \mathbf{y}})$$

where

$$\mathbf{m}_{Y_0 | \mathbf{Y} = \mathbf{y}} = \mathbf{m}_0 + \mathbf{k}^T \Sigma^{-1} (\mathbf{y} - \mathbf{m})$$

$$\Sigma_{Y_0 | \mathbf{Y} = \mathbf{y}} = \Sigma_0 - \mathbf{k}^T \Sigma^{-1} \mathbf{k}$$

Question: what if we don't know $m(\mathbf{s}; \boldsymbol{\beta}), c(h; \boldsymbol{\theta})$?

⇒ The mean and covariance functions are typically unknown and must be estimated from the data $\mathbf{y}$. This will be covered in the next module

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These slides cover:

- The problem of **spatial interpolation**
- **Stationarity** and **Isotropy** of a spatial process
- **Gaussian Process Spatial Models**

R functions/tricks to know:

- `vgram` (under the package `fields`) for visualizing spatial dependence
- `image.plot` (under the package `fields`) for visualizing spatial images
- Some matrix calculation tricks for speeding up computation

Bochner's Theorem

A complex-valued function C on $\mathbb{R}^d$ is the covariance function for a weakly stationary mean square continuous complex-valued random process on $\mathbb{R}^d$ if and only if it can be represented as

$$C(\mathbf{h}) = \int_{\mathbb{R}^d} \exp(i\omega^T \mathbf{h}) F(d\omega),$$

with F a positive finite measure. When F has a density with respect to Lebesgue measure, we have the spectral density f and

$$f(\omega) = \frac{1}{2\pi} \int_{\mathbb{R}^d} \exp(-i\omega^T \mathbf{h}) C(\mathbf{h}) d\mathbf{h}$$

Practical Takeaway

Bochner's theorem establishes a one-to-one relationship between stationary covariance functions and spectral densities. It provides a powerful tool for constructing valid covariance models and understanding spatial dependence across different spatial scales.